Smith normal form of the n-cycle class sum

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Abstract

Let M_n be the integer matrix of multiplication by the sum of all n-cycles on the center of $\mathbb{Z}S_n$, in the conjugacy-class-sum basis. We determine its Smith normal form for every positive integer n. It has $p(n) - n$ zero entries and nonzero invariant factors $d_0 = 1$ and $d_r = a_r r!/\gcd(n, r)$ for $1 \le r < n$, where $a_r = 2$ when $r \equiv n \pmod 2$ and $a_r = 1$ otherwise, and $p(n)$ is the partition number. This confirms Stanley's two conjectures for odd primes and twice odd primes, posed in 2019, when he also suggested that further data might lead to a conjecture for all n. Starting from the classical hook-character expression for the structure constants, we determine two integral coefficient lattices and reduce the induced pairing to an anti-triangular matrix with rowwise divisibility. The image-lattice theorem and the divisor chain are formalized in Lean from group-ring definitions, including a proof of the character expression. The equivalent diagonal form and the prime cases follow by the written basis-change and specialization arguments.

Checked in Lean from the definitions; not yet reviewed by experts.

The proofs, the Lean formalization and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 Introduction

For a partition $\mu \vdash n$, write K_μ for the sum of the permutations of cycle type μ in $\mathbb{Z}S_n$. The elements K_μ form a $\mathbb{Z}$-basis of the center $Z(\mathbb{Z}S_n)$, identifying it with $\mathbb{Z}^{p(n)}$, where $p(n)$ is the number of partitions of n. We consider the matrix M_n defined by

$$K_{(n)}K_\nu = \sum_{\mu \vdash n} M_n[\mu, \nu]K_\mu. \tag{1}$$

Thus the rows record output cycle types and the columns record input cycle types. The Smith normal form of an integer matrix A is a diagonal matrix obtained by invertible integral row and column operations, with positive nonzero entries $d_0 \mid d_1 \mid \cdots$ followed by zeros. Equivalently, it describes the embedding of the column lattice of A in its ambient free abelian group up to integral changes of basis.

Stanley asked for the Smith normal forms of class-sum multiplication matrices in 2019 [9], inspired by Lindzey's question about the Cayley graph generated by n-cycles [5]. Lindzey considers the full $n!$-dimensional group algebra; here the operator is restricted to its center. For the n-cycle class Stanley gave data, observed that the rank is n, and conjectured explicit formulas when n is an odd prime and when it is twice an odd prime. He suggested that more data might lead to a conjecture for all n. His broader question concerns arbitrary conjugacy classes and finite groups. The following theorem determines the n-cycle case for every n.

Theorem 1.1. *Let $n \geq 1$, and let $p(n)$ denote the number of partitions of n. Define*

$$d_0 = 1, \qquad d_r = \frac{a_r r!}{\gcd(n, r)} \quad (1 \leq r < n), \qquad a_r = \begin{cases} 2 & r \equiv n \pmod 2, \\ 1 & r \not\equiv n \pmod 2. \end{cases} \tag{2}$$

These are positive integers satisfying $d_r \mid d_{r+1}$ for $0 \leq r < n-1$. There is a basis

$$b_0, \ldots, b_{n-1}, b'_1, \ldots, b'_{p(n)-n}$$

of $\mathbb{Z}^{p(n)}$ such that

$$M_n \mathbb{Z}^{p(n)} = \bigoplus_{r=0}^{n-1} d_r \mathbb{Z} b_r. \tag{3}$$

Consequently, $\mathrm{SNF}(M_n) = \mathrm{diag}(d_0, \ldots, d_{n-1}, 0, \ldots, 0)$, *with $p(n) - n$ zeros.*

For an odd prime $n = p$, the factors are $r!$ at even indices and $2r!$ at odd indices; for $n = 2p$, they are the factorials with $p!$ replaced by $(p-1)!$. Thus both of Stanley's conjectures follow (Corollary 7.1). Section 7 compares the formula with his examples, including a discrepancy at index six in the displayed $n = 12$ list.

The proof starts with the hook-character generating polynomials in q, substitutes $q = -x$, and passes to coefficients in the shifted variable $y = x - 1$. We determine the integral lattice spanned by these coefficient vectors and the lattice obtained by dividing each vector by the corresponding centralizer order. In these coordinates the classical character pairing is anti-triangular. Bases for the two lattices expose its anti-diagonal entries, and row divisibility then identifies the Smith factors.

The character-theoretic study of factorizations into n-cycles includes Stanley's work [7]. Goupil and Schaeffer [3] give an explicit formula for factoring an n-cycle into two permutations with prescribed cycle types, and Biane [1] gives an algebraic proof of that formula. These connection coefficients are related to the entries of M_n by the normalization stated in Section 2. The present proof determines the integral image lattice of this matrix.

Stanley's survey [8] gives the broader context of Smith forms in combinatorics. Cai and Stanley [2] determine the Smith form of the up–down operator on Young's lattice, equivalently a differential operator on symmetric functions. That operator provides related context for (1). The representation-theoretic ingredients below are classical; see, for example, Sagan [6]. Grinberg's recent exposition [4] provides a detailed treatment of class sums and symmetric-group representations, and was a source for this background.

The Lean development proves (3) and the divisor chain from the definitions of permutations and class sums, including a proof of the character formula. The passage from (3) to diagonal Smith form and the prime specializations are written deductions given below. Section 8 states the formal theorem precisely and explains its connection with the written proof.

2 The character formula and a change of basis

Fix $n \geq 1$ and put $N = n - 1$. For $\mu \vdash n$, let $\ell(\mu)$ be its number of parts and $m_j(\mu)$ the multiplicity of j. Set

$$z_\mu = \prod_{j \geq 1} j^{m_j(\mu)} m_j(\mu)!, \qquad \epsilon_\mu = (-1)^{\ell(\mu)-1}.$$

The number z_μ is the order of the centralizer of a permutation of type μ. Write $\chi_k = \chi^{(n-k,1^k)}$ for the hook character, where $0 \leq k \leq N$. The hook characters are the characters of the exterior powers

of the standard representation, so

$$\sum_{k=0}^{N} \chi_k(\mu) q^k = \frac{\prod_{j=1}^{\ell(\mu)} (1-(-q)^{\mu_j})}{1+q}. \tag{4}$$

Indeed, the characteristic polynomial on the permutation representation is the product of the characteristic polynomials of its cycles. Removing its one-dimensional trivial summand divides the exterior-power generating function by $1+q$.

Proposition 2.1. *For $\mu, \nu \vdash n$,*

$$M_n[\mu,\nu] = \frac{1}{z_\nu} \sum_{k=0}^{N} (-1)^k k!(N-k)! \chi_k(\mu)\chi_k(\nu). \tag{5}$$

Proof. Frobenius's class-multiplication formula gives

$$M_n[\mu,\nu] = \frac{|K_{(n)}|\,|K_\nu|}{n!} \sum_{\lambda \vdash n} \frac{\chi^\lambda((n))\chi^\lambda(\nu)\chi^\lambda(\mu)}{\chi^\lambda(1)}.$$

All symmetric-group characters here are real, and every conjugacy class is closed under inversion. By the Murnaghan–Nakayama rule, only hooks contribute: $\chi_k((n)) = (-1)^k$. Their dimensions are $\binom{N}{k}$, while $|K_{(n)}| = N!$ and $|K_\nu| = n!/z_\nu$. Substitution gives (5). □

To relate the normalization to the factorization literature, let $c_{\mu\nu}^{(n)}$ count the factorizations of a fixed n-cycle as a product of permutations of types μ and ν. Equation (2) of Biane [1], identified there with equation (4) of Goupil and Schaeffer [3], gives

$$c_{\mu\nu}^{(n)} = \frac{n}{z_\mu z_\nu} \sum_{k=0}^{N} (-1)^k k!(N-k)! \chi_k(\mu)\chi_k(\nu).$$

Thus $M_n[\mu,\nu] = (z_\mu/n) c_{\mu\nu}^{(n)}$. The multiplier depends on the row and can be nonintegral; this normalization identity is not an equivalence under unimodular operations. The character expression supplies the entries, while the lattice arguments below determine their integral invariant factors.

Introduce polynomials

$$H_\mu(y) = \prod_{j=1}^{\ell(\mu)} ((1+y)^{\mu_j} - 1), \qquad \eta_\mu(i) = [y^{i+1}] H_\mu(y), \quad 0 \le i \le N,$$

$$\Phi_\mu(y) = \frac{H_\mu(y)}{z_\mu}, \qquad e_s = s!\,\gcd(n, s+1), \quad 0 \le s \le N.$$

The order of H_μ is $\ell(\mu)$, and its lowest coefficient is $\prod_j \mu_j$. For $0 \le i, s \le N$, define

$$Q(i,s) = \begin{cases} (-1)^{i+s} \dfrac{(N-i)!(N-s)!}{(N-i-s)!}, & i+s \le N, \\ 0, & i+s > N. \end{cases} \tag{6}$$

Lemma 2.2. *For $0 \le i, s \le N$,*

$$\sum_{k=0}^{N} (-1)^k k!(N-k)! [x^k](x-1)^i\,[x^k](x-1)^s = Q(i,s).$$

Consequently,

$$M_n[\mu,\nu] = \frac{\epsilon_\mu \epsilon_\nu}{z_\nu} \sum_{i,s=0}^{N} \eta_\mu(i) Q(i,s) \eta_\nu(s). \tag{7}$$

Proof. The coefficient $[x^k](x-1)^i$ is $(-1)^{i-k}\binom{i}{k}$. Using $k!\binom{i}{k}(N-k)! = i!(N-i)!\binom{N-k}{N-i}$, the sum in the lemma becomes

$$(-1)^{i+s} i!(N-i)! \sum_{k=0}^{s} (-1)^k \binom{s}{k} \binom{N-k}{N-i}.$$

Pascal's identity, applied successively s times, gives

$$\sum_{k=0}^{s} (-1)^k \binom{s}{k} \binom{N-k}{m} = \binom{N-s}{m-s},$$

where an out-of-range lower index means zero. Take $m = N-i$ and simplify factorials to obtain (6). Now set $g_\mu(x) = \prod_j (1-x^{\mu_j})/(1-x)$. Equation (4) implies $\chi_k(\mu) = (-1)^k [x^k] g_\mu(x)$, and substitution gives

$$g_\mu(1+y) = \epsilon_\mu H_\mu(y)/y = \epsilon_\mu \sum_i \eta_\mu(i) y^i.$$

Apply the first assertion to (5). □

3 The normalized coefficient lattice

Define the rational vector $\alpha(\nu)$ by

$$\alpha(\nu)_s = \epsilon_\nu e_s [y^{s+1}] \Phi_\nu = \frac{\epsilon_\nu e_s \eta_\nu(s)}{z_\nu}.$$

Lemma 3.1. *The vectors $\alpha(\nu)$, for $\nu \vdash n$, belong to $\mathbb{Z}^n$ and span $\mathbb{Z}^n$ over $\mathbb{Z}$.*

Proof. We first prove integrality. Allow μ to be any multiset of positive integers, including the empty multiset, and put

$$\phi_j(y) = \frac{(1+y)^j - 1}{j}, \qquad \Phi_\mu = \prod_{j \geq 1} \frac{\phi_j^{m_j(\mu)}}{m_j(\mu)!}, \qquad I_t(\mu) = t![y^t]\Phi_\mu.$$

For the empty multiset, $\Phi_\varnothing = 1$. Removing one part j is denoted by $\mu - j$. Since $\phi_j' = (1+y)^{j-1}$,

$$\Phi_\mu' = \sum_{j : m_j(\mu) > 0} (1+y)^{j-1} \Phi_{\mu - j}. \tag{8}$$

Induction on the sum of the parts proves $I_t(\mu) \in \mathbb{Z}$ for every $t \geq 0$. The constant coefficient is immediate; for $t \geq 1$, (8) expresses $I_t(\mu)$ as a sum of

$$\binom{j-1}{i} \frac{(t-1)!}{(t-1-i)!} I_{t-1-i}(\mu - j), \qquad 0 \leq i \leq \min(j-1, t-1),$$

each an integer by induction.

When the sum of the parts is n, the identity $jm_j(\mu)\Phi_\mu = ((1+y)^j - 1)\Phi_{\mu-j}$ yields

$$n\Phi_\mu = \sum_{j:m_j(\mu)>0} ((1+y)^j - 1)\Phi_{\mu-j}. \tag{9}$$

For $t \geq 1$, multiply the coefficient of y^t by $(t-1)!$. The right side is a sum of

$$\binom{j}{i}\frac{(t-1)!}{(t-i)!}I_{t-i}(\mu - j), \qquad 1 \leq i \leq \min(j,t),$$

so $nI_t(\mu)/t$ is an integer. Bézout's identity for n and t then shows that $\gcd(n,t)I_t(\mu)/t$ is an integer. Setting $t = s+1$ proves the required integrality.

For spanning, fix a length $\ell = s+1$. The vectors from partitions of this length vanish in coordinates below s, and their sth coordinates, up to sign, are

$$\frac{(\ell-1)!\gcd(n,\ell)}{\prod_j m_j(\nu)!}. \tag{10}$$

We show that the gcd of these integers is one. If $\ell = n$, the partition 1^n gives one. Suppose $\ell < n$, fix a prime p, and write $v = v_p(\gcd(n,\ell))$. If $v = 0$, the partition $(n-\ell+1, 1^{\ell-1})$ gives $\gcd(n,\ell)$, prime to p. If $v > 0$, put $q = p^v$ and $A = (n-\ell)/q + 1 \geq 2$. The partition $(A^q, 1^{\ell-q})$ gives

$$\binom{\ell-1}{q-1}\frac{\gcd(n,\ell)}{q}.$$

The second factor is prime to p. In $\binom{\ell-1}{q-1} = \prod_{i=1}^{q-1}(\ell-i)/i$, the valuations of numerator and denominator agree term by term, because $v_p(i) < v \leq v_p(\ell)$. Thus the first factor is also prime to p. No prime divides every number in (10). Bézout therefore gives a combination f_s of the length-$s+1$ vectors with coordinate one at s and zero below it. The vectors $f_0, \ldots, f_N$ form a unitriangular integral matrix, hence a basis of $\mathbb{Z}^n$. □

4 The integral coefficient lattice

Let U be the $\mathbb{Z}$-span of the vectors η_μ in $\mathbb{Z}^n$. Extend the notation (2) by setting $a_0 = n$, so that $d_r = a_r r!/\gcd(n,r)$ also holds at $r = 0$, using $\gcd(n,0) = n$.

Lemma 4.1. *The lattice U has a basis $u_0, \ldots, u_N$ such that $u_r(i) = 0$ for $i < r$ and $u_r(r) = a_r$. Each u_r is an integral combination of the η_μ with $\ell(\mu) = r+1$.*

Proof. Take $u_0 = \eta_{(n)}$. When $N > 0$, take $u_N = \eta_{1^n}$. For $1 \leq r < N$, the two partitions

$$(n-r, 1^r), \qquad (n-r-1, 2, 1^{r-1})$$

have length $r+1$ and leading coefficients $n-r$ and $2(n-r-1)$. Their gcd is $\gcd(n-r,2) = a_r$, so Bézout gives the desired u_r. This constructs independent vectors with the asserted leading terms.

We prove that they span. Identify $v \in \mathbb{Z}^n$ with $\sum_{i=0}^N v(i)y^i$. Every $v \in U$ satisfies $n \mid v(0)$. It also satisfies

$$T_N(v) \equiv v \pmod 2, \qquad T_m(f) = (1+y)^m f(y/(1+y)). \tag{11}$$

Indeed, $T_{a+b}(fg) = T_a(f)T_b(g)$ and

$$T_m((1+y)^m - 1) = (1+2y)^m - (1+y)^m \equiv (1+y)^m - 1 \pmod 2.$$

Consequently $T_n(H_\mu) \equiv H_\mu \pmod 2$. Since $T_n(H_\mu) = yT_N(H_\mu/y)$, cancellation of y proves (11) on the generators and hence on U.

Suppose $v(i) = 0$ for $i < r$. At $r = 0$ we have $a_0 \mid v(0)$. For $1 \le r < N$, comparing coefficients of y^{r+1} in (11) gives $(N-r)v(r) \equiv 0 \pmod 2$. When $r \equiv n \pmod 2$, the factor $N-r$ is odd, so $a_r = 2$ divides $v(r)$. In all other cases, including $r = N$, $a_r = 1$. Subtracting $(v(r)/a_r)u_r$, successively for $r = 0, \dots, N$, therefore expresses every element of U in the constructed basis. □

5 Divisibility of the pairing

For a partition μ and $0 \le s \le N$, put

$$\Gamma(\mu, s) = \frac{1}{e_s} \sum_{i=0}^{N} \eta_\mu(i) Q(i, s).$$

Lemma 5.1. *If $\ell(\mu) = r + 1$, then $\Gamma(\mu, s) \in d_r\mathbb{Z}$ for every s.*

Proof. Rewrite (6) as

$$Q(i, s) = (-1)^{i+s} s!(N - s)! \binom{N - i}{s}.$$

Thus, for the integer $X_s = \sum_i (-1)^i \binom{N-i}{s} \eta_\mu(i)$,

$$\Gamma(\mu, s) = (-1)^s \frac{(N - s)!}{\gcd(n, s + 1)} X_s. \tag{12}$$

Terms vanish unless $r \le i \le N - s$. If $s > N - r$ the assertion follows at once. If $s = N - r$, then $X_s = (-1)^r \prod_j \mu_j$ and

$$\Gamma(\mu, N - r) = \pm \frac{r!}{\gcd(n, r)} \prod_j \mu_j.$$

The product is divisible by a_r: for $r = 0$ it equals n; for $a_r = 2$ and $r \ge 1$, $r + 1$ odd parts would have sum of parity $r + 1$, contrary to $n \equiv r \pmod 2$.

It remains to treat $s < N - r$. Put $t = N - s > r$ and $b = \gcd(n, s + 1)$. As $b \mid n - (s + 1) = t$, we have

$$\frac{t!}{b} = r!R, \qquad R = (r + 1)(r + 2) \cdots (t - 1)\frac{t}{b} \in \mathbb{Z}.$$

It suffices to show $a_r \mid \gcd(n, r)R$. This is immediate if $r = 0$, or $a_r = 1$, or n, r are even. The remaining case has $a_r = 2$ and n, r odd. Then b is odd. If $t > r + 1$, R contains the even factor $r + 1$; if $t = r + 1$, its factor t/b is even. This proves the required divisibility, including integrality. □

6 Assembly and the Smith form

For $0 \le r, s \le N$, define

$$G(r, s) = \frac{1}{e_s} \sum_i u_r(i) Q(i, s).$$

Lemma 5.1 and the length restriction in Lemma 4.1 show that d_r divides every entry of row r. Moreover $G(r, s) = 0$ if $r + s > N$, and

$$G(r, N - r) = \frac{a_r Q(r, N - r)}{e_{N-r}} = (-1)^N \frac{a_r r!}{\gcd(n, n - r)} = (-1)^N d_r.$$

Thus

$$G = \operatorname{diag}(d_0, \dots, d_N)V, \tag{13}$$

where V is an integral anti-triangular matrix with entries ± 1 on the anti-diagonal. Its determinant is ± 1, so V is unimodular.

Write $\eta_\mu = \sum_r X(\mu, r)u_r$ using Lemma 4.1, and write $u_r = \sum_\mu W(r, \mu)\eta_\mu$ using the construction there. The matrices X and W are integral, and independence of the u_r gives $WX = I_n$. Let $E = \operatorname{diag}(\epsilon_\mu)$ and let α be the matrix whose columns are $\alpha(\nu)$. Equation (7) and the definitions give

$$M_n = E\Gamma\alpha = EXG\alpha.$$

By Lemma 3.1, $\alpha : \mathbb{Z}^{p(n)} \to \mathbb{Z}^n$ is surjective. Since V is unimodular, (13) now yields

$$M_n\mathbb{Z}^{p(n)} = EX \operatorname{diag}(d_0, \dots, d_N)\mathbb{Z}^n = \bigoplus_{r=0}^{N} d_r\mathbb{Z}b_r, \qquad b_r = EX\mathbf{e}_r.$$

The map WE is a left inverse of EX, because $E^2 = I$. Therefore

$$\mathbb{Z}^{p(n)} = EX\mathbb{Z}^n \oplus \ker(WE).$$

The kernel is a free abelian group, being a subgroup of a finitely generated free abelian group. Choosing its basis extends the b_r to a basis of $\mathbb{Z}^{p(n)}$, proving (3).

For completeness, all d_r are positive integers: $d_0 = 1$, and for $r \geq 1$ the divisor $\gcd(n, r)$ divides $r!$. We also have $d_0 \mid d_1$ whenever the latter occurs. For $1 \leq r < N$,

$$\frac{d_{r+1}}{d_r} = \frac{a_{r+1}}{a_r}\frac{(r+1)\gcd(n, r)}{\gcd(n, r+1)}.$$

Since $\gcd(n, r+1) \mid r+1$, only $a_r = 2$ could cause a denominator. In that case $a_{r+1} = 1$. If n, r are even, $\gcd(n, r)$ is even and $\gcd(n, r+1)$ is odd. If n, r are odd, $r+1$ is even and $\gcd(n, r+1)$ is odd. The ratio is integral in both cases, establishing the divisor chain.

Finally, choose an integral change of basis P carrying b_r to the first n standard basis vectors. Then PM_n maps $\mathbb{Z}^{p(n)}$ onto $\bigoplus_r d_r\mathbb{Z}\mathbf{e}_r$. Lift the basis vectors $d_r\mathbf{e}_r$ of this image to the domain. These lifts, together with a basis of the kernel, form a basis of the domain: the map onto its free image splits. The corresponding unimodular column change Q satisfies

$$PM_nQ = \operatorname{diag}(d_0, \dots, d_{n-1}, 0, \dots, 0).$$

The image has rank n, so precisely $p(n) - n$ entries vanish. This proves Theorem 1.1.

7 Stanley's conjectures and examples

Corollary 7.1. *If $n = p$ is an odd prime, the nonzero invariant factor at index r is $r!$ for even r and $2r!$ for odd r, for $0 \leq r < p$. If $n = 2p$ for an odd prime p, the list is $0!, 1!, \dots, (2p-1)!$ with $p!$ replaced by $(p-1)!$.*

Proof. For $1 \leq r < p$, $\gcd(p, r) = 1$, and $r \equiv p \pmod 2$ means r is odd. This proves the first assertion, including $r = 0$ separately. When $n = 2p$, even r with $0 < r < 2p$ have gcd two and $a_r = 2$. Odd $r \neq p$ have gcd one and $a_r = 1$. At $r = p$ the factor is $p!/p = (p-1)!$. Again $d_0 = 1$. □

The theorem also gives

$$Z(\mathbb{Z}S_n)/K_{(n)}Z(\mathbb{Z}S_n) \cong \mathbb{Z}^{p(n)-n} \oplus \bigoplus_{r=0}^{n-1} \mathbb{Z}/d_r\mathbb{Z}.$$

These corollaries are immediate written consequences of the lattice statement; separate Lean declarations for them are not needed for the formal theorem in Section 8.

For $n = 1$, the matrix is $[1]$. The nonzero factors for $n = 2, 3, 4, 5$ are respectively

$$(1, 1), \quad (1, 2, 2), \quad (1, 1, 2, 6), \quad (1, 2, 2, 12, 24).$$

The formula at $n = 9$ gives

$$0!,\ 2 \cdot 1!,\ 2!,\ \tfrac{2}{3}\,3!,\ 4!,\ 2 \cdot 5!,\ \tfrac{1}{3}\,6!,\ 2 \cdot 7!,\ 8!,$$

agreeing term by term with Stanley's example [9]. At $n = 12$, it gives

$$0!,\ 1!,\ 2!,\ \tfrac{1}{3}\,3!,\ \tfrac{1}{2}\,4!,\ 5!,\ \tfrac{1}{3}\,6!,\ 7!,\ \tfrac{1}{2}\,8!,\ \tfrac{1}{3}\,9!,\ 10!,\ 11!.$$

The entry at index six is $6!/3$. The displayed $n = 12$ example in [9] instead has $2 \cdot 6!$ at this position; that entry does not divide the following $7!$, so the displayed entries do not form a divisibility chain. Our formula gives $6!/3$ instead. The other entries agree. These examples illustrate the formula and are not inputs to its proof.

8 Formalization and reproducibility

The accompanying Lean project defines S_n as the permutations of $\{0, \ldots, n-1\}$ and works in its integral monoid algebra. The class sum includes each permutation of the prescribed cycle type with coefficient one. The matrix entry is the coefficient of a chosen permutation of type μ in $K_{(n)}K_\nu$; the development proves that the choice has the required cycle type and that these entries give the expansion (1).

In ASCII Lean notation, the final theorem is:

```
theorem stanley_class_sum_snf (n : Nat) (hn : 0 < n) :
    And (HasStanleySNF n (M n))
      (forall r, r + 1 < n ->
        Dvd.dvd (stanleyD n r) (stanleyD n (r + 1)))
```

Here `stanleyD` is (2), with its division proved exact. The predicate `HasStanleySNF` asserts the existence of a finite index set κ and an integral basis indexed by $\{0, \ldots, n-1\} \sqcup \kappa$, for which the image is the span of $d_r b_r$. This is precisely (3). Positivity and independence turn that span into the direct sum stated in the theorem. The written basis-change argument in Section 6 identifies it with the usual diagonal Smith form. In particular the formal statement does not replace the group-ring matrix by a matrix defined through a character formula.

For the character formula, the formal proof uses an explicit model of the exterior algebra of $\mathbb{Q}^n$, with basis indexed by subsets. Permutations act by signed permutations of these basis vectors. Exterior multiplication by $e = e_1 + \cdots + e_n$ and contraction by the all-ones linear functional give operators E and I satisfying $IE + EI = n$. Thus $P = IE/n$ is an equivariant projection onto $\ker I$. Taking traces of the degree-k action on this image gives the coefficients in (4).

For the interior degrees $1 \le k < n$, the degree-k commutant is determined by the signed action on pairs of k-subsets: a transposition in their difference kills an entry when their intersection has size at most $k-2$, and the remaining orbits leave a two-dimensional span of the identity and the projection P. Restriction to the image of P makes every commuting operator scalar. At degree 0, the representation is one-dimensional and P is the identity, so the same scalar conclusion holds directly. Taking traces of conjugacy averages consequently gives

$$\sum_{g \in S_n} \chi_k(agbg^{-1}) = \frac{n!}{\binom{N}{k}} \chi_k(a)\chi_k(b).$$

Evaluation of (4) at $q = -1$ gives the indicator identity $\sum_k (-1)^k \chi_k(\mu) = n\,\mathbf{1}_{\mu=(n)}$. This identity, the preceding conjugacy average, and orbit–stabilizer prove Proposition 2.1 directly in Lean. Consequently, the formal proof does not take the classical representation-theoretic theorems used in its shorter written derivation as extra hypotheses.

The remaining development follows the polynomial pairing, the two lattice lemmas, row divisibility, and assembly above. It uses Lean's kernel and the mathlib library; no certificate generator, numerical approximation, or external computational oracle is required. From the bundle's `lean/` directory, the standard commands are

```
lake exe cache get
lake build
```

The build includes checks of the final statement and its logical dependencies. The accompanying Lean guide documents these checks and the correspondence of definitions. The paper can be rebuilt from its source with `pdflatex`.

Acknowledgments

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